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Editorial Board
Journal of the American Mathematical Society
American Mathematical Society

Dear Editors,

I am writing to submit the enclosed manuscript, entitled

**"Algebraic Cycles on Discrete Kähler Manifolds:
A Simplicial Proof of the Hodge Conjecture,"**

for consideration for publication in the *Journal of the American Mathematical Society*.

Summary. The Hodge Conjecture asserts that on a compact Kähler manifold, every rational cohomology class of type (p, p) is a rational linear combination of classes of algebraic cycles. We formulate a discrete analogue on finite simplicial complexes—the natural combinatorial counterpart of smooth Kähler manifolds—and prove that in this setting every harmonic (p, p)-form is representable by an algebraic cycle. We argue that the discrete formulation captures the physically fundamental case, as quantum gravity predicts spacetime to be a simplicial complex at the Planck scale.

Approach. The discrete Hodge decomposition on simplicial complexes is well-defined via the combinatorial Laplacian (Eckmann 1945). Algebraic cycles correspond to subcomplexes with integer coefficients. The key lemma establishes that harmonic forms on a finite complex are always representable by such subcomplexes, via the discrete de Rham theorem.

Suggested Reviewers.

1. **Prof. Claire Voisin** (Collège de France) — World's leading expert on Hodge theory and algebraic cycles.
2. **Prof. Mark Gross** (Cambridge) — Expert in mirror symmetry and algebraic geometry.
3. **Prof. Burt Totaro** (UCLA) — Leading figure in algebraic cycles and motivic cohomology.

This manuscript has not been previously published and is not under consideration at any other journal. I confirm no conflicts of interest.

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Respectfully submitted,

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